

WEEK - 16

Challenge 1: Build a Single Neuron Classifier

Understand how a **perceptron (single neuron)** makes decisions. No need for deep learning libraries

Create a Python program that:

- Takes two inputs: x_1 , x_2
- Computes: z , output

Challenge 2: Try Different Activation Functions

Write a Python script that:

- Implements:
 - Sigmoid
 - Tanh
 - ReLU
- For input values: $x = [-5, -2, -1, 0, 1, 2, 5]$

Print outputs of each activation. Plot graphs of the three activations.